AEC
Algerian Engineering Competition

Optimization of Aerial Firefighting Interventions Using Artificial Intelligence

for :

طاسيلي للعمل الجوي
Tassili Travail Aérien

team robobo

General Introduction

In recent years, aerial robotics—especially autonomous drones (UAVs)—has become a rapidly growing field due to advancements in robotics, electronics, and miniaturization. These drones are now widely used in various sectors like mapping, agriculture, military, environmental monitoring, and even delivery services. A key challenge is enabling drones to follow flight plans autonomously while avoiding obstacles. In this project, such a drone is used for early forest fire detection. It flies over forests, takes photos, and sends them to a processing center where an AI system analyzes the images to detect signs of fire

The problem

Wildfire Response Time is Measured in Destructionm

- The single most critical factor in wildfire containment is response time. Current methods relying on watchtowers or public reporting introduce fatal delays.
- This latency between Ignition, Detection, Verification, and Intervention often takes hours, not minutes.
- This delay allows small, manageable fires to escalate into uncontrollable disasters, causing devastating ecological and economic damage.

Our Goal: To create a system that detects and verifies fires in minutes, not hours.

AEC
Algerian Engineering Competition

Our Solution

Project Aegis is an intelligent, two-tier system designed to reduce the detection-to-verification time to under 10 minutes.

- "The Tripwire": A strategically deployed network of low-power IoT sensors (thermal, smoke, gas) provides 7/24 monitoring in high-risk zones, acting as an early warning layer..

- "The Eyes in the Sky": A fleet of AI-powered autonomous drones is stationed at fixed depots. When a tripwire is triggered, a drone is dispatched in real-time to investigate and provide live, actionable intelligence.

AEC
Algerian Engineering Competition

The Autonomous Loop in Action

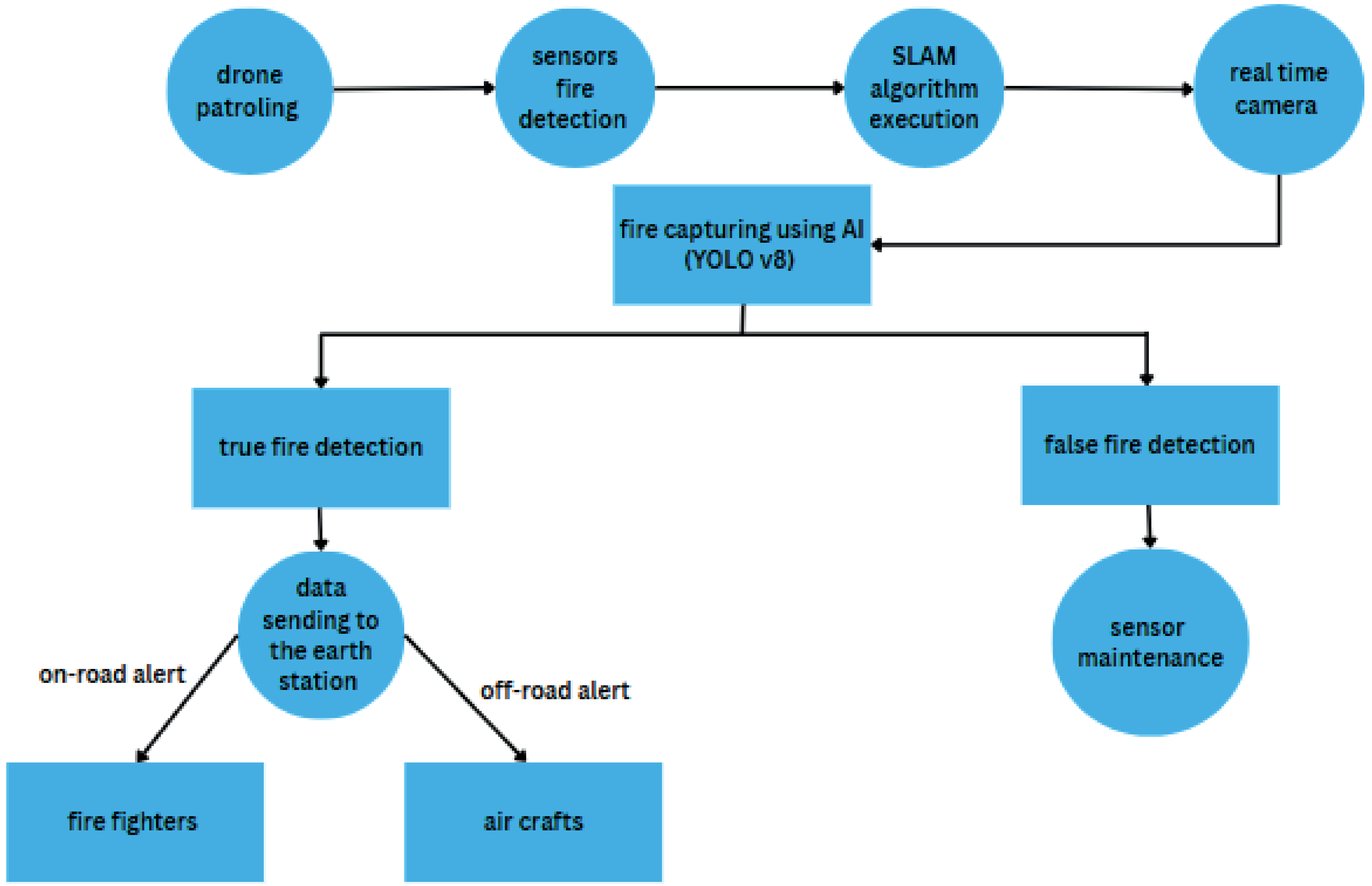
1. **Detection (Tripwire Triggered)**: A static sensor detects an anomaly (heat/smoke) and transmits a Level 1 (Unverified) Alert with GPS coordinates to the central server.

2. **Triage & Dispatch (AI in Action)**: The central system calculates and dispatches the optimal drone (based on proximity, battery, and status) to the alert location.

3. **Verification (Eye in the Sky):** The drone arrives on-site, executes a pre-programmed verification routine, and uses onboard AI (YOLOv8) to analyze thermal and visual feeds to confirm the fire's presence.

4. **Escalation (Actionable Intelligence):** If confirmed, a Level 2 (Verified) Alert is automatically escalated to emergency services, complete with GPS coordinates, live video feed, estimated fire size, and access routes.

Real-Time Operational Loop



The Platform: Our Chosen Drone Architecture

The Technology: A Platform Built for the Challenge

- **Platform Type:** Hybrid VTOL (Vertical Take-Off and Landing) quadcopter design. This merges the hovering and maneuverability of a quadcopter with the endurance and high-speed performance of a fixed-wing aircraft.
- **Performance Specifications:**

  Endurance: 120–90 minutes of flight time.

  Maximum Speed: Up to 120 km/h in fixed-wing mode.

  Wind Resistance: Stable in winds up to 60 km/h.

- **Onboard Intelligence:** NVIDIA Jetson Nano or Raspberry Pi 4 for GPU-accelerated, real-time AI inference.
- **Sensor Payload:** Dual-camera system featuring a FLIR Lepton 3.5 thermal camera and a high-resolution Raspberry Pi HQ visual camera for multi-modal verification

Onboard Fire Detection with YOLOv8

Real-Time Fire Detection

Method: We use YOLOv8, a state-of-the-art, real-time object detection algorithm ideal for onboard drone deployment due to its exceptional speed and accuracy.

Training: Our model was trained for 20 epochs on a curated dataset from Roboflow, containing over 1,000 images of fire, smoke, and diverse background scenes. This resulted in a robust model achieving a mAP@0.5 of 0.58, a strong result for an early-stage model.

Deployment: The trained model (best.pt) runs directly on the drone's onboard computer (Jetson/Pi), enabling it to instantly identify and localize "fire" and "smoke" from its live video stream..

Dataset and Model Training

Model Integration

```
[ ] from ultralytics import YOLO

    model = YOLO('yolov8n.pt')
    model.train(data='/content/dataset/data.yaml', epochs=20, imgsz=640)
```

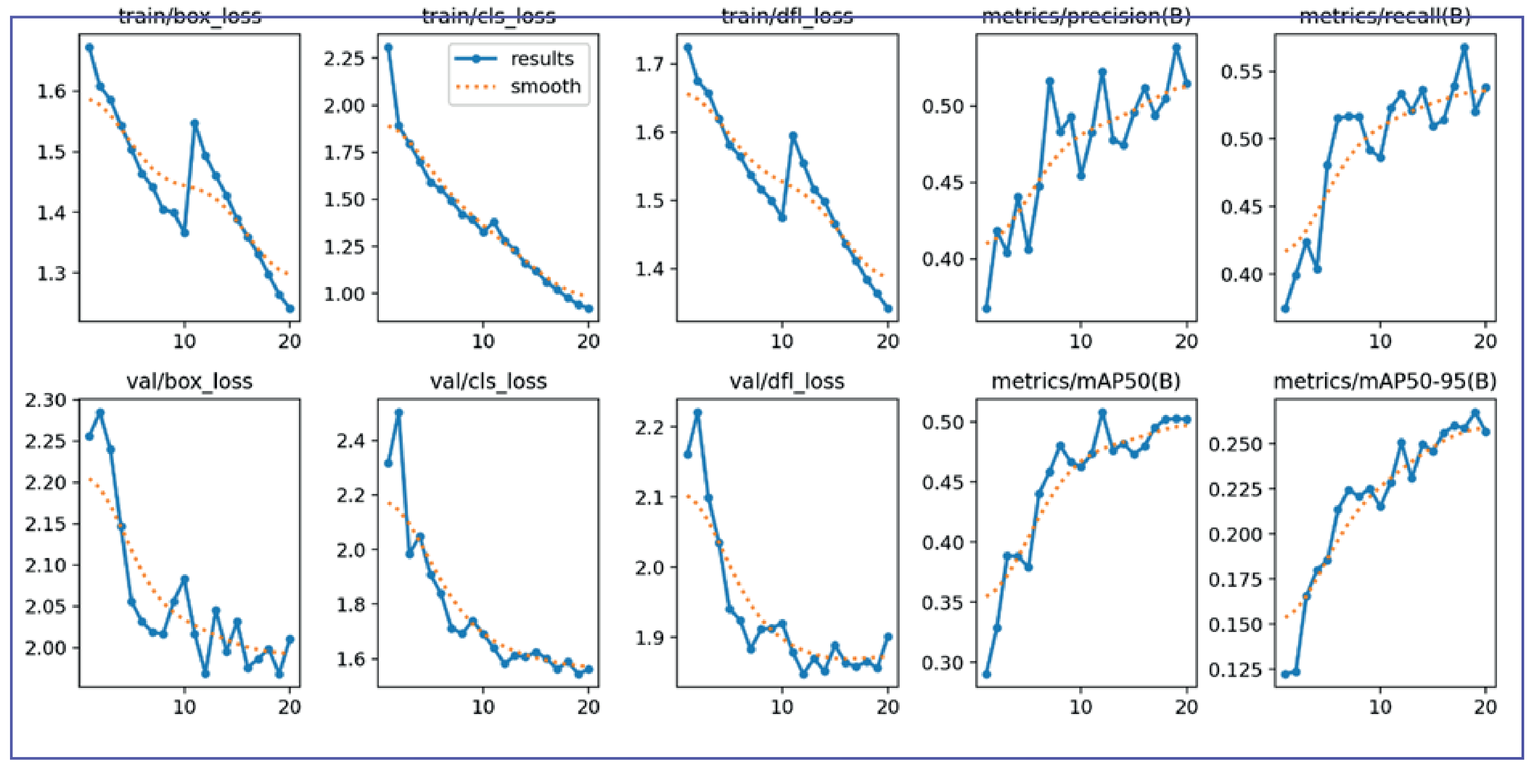
```
from ultralytics import YOLO

model = YOLO('/content/runs/detect/train2/weights/best.pt')

[ ] results = model.predict(source='/content/dataset/test/images/fire-92-_png.rf.35093d5b2aacdc5ba0ba65b5b7cb6146.jpg', save=True, conf=0.25)

image 1/1 /content/dataset/test/images/fire-92-_png.rf.35093d5b2aacdc5ba0ba65b5b7cb6146.jpg: 640x640 6 fires, 3 smokes, 8.9ms
Speed: 2.4ms preprocess, 8.9ms inference, 1.7ms postprocess per image at shape (1, 3, 640, 640)
Results saved to runs/detect/predict2
```

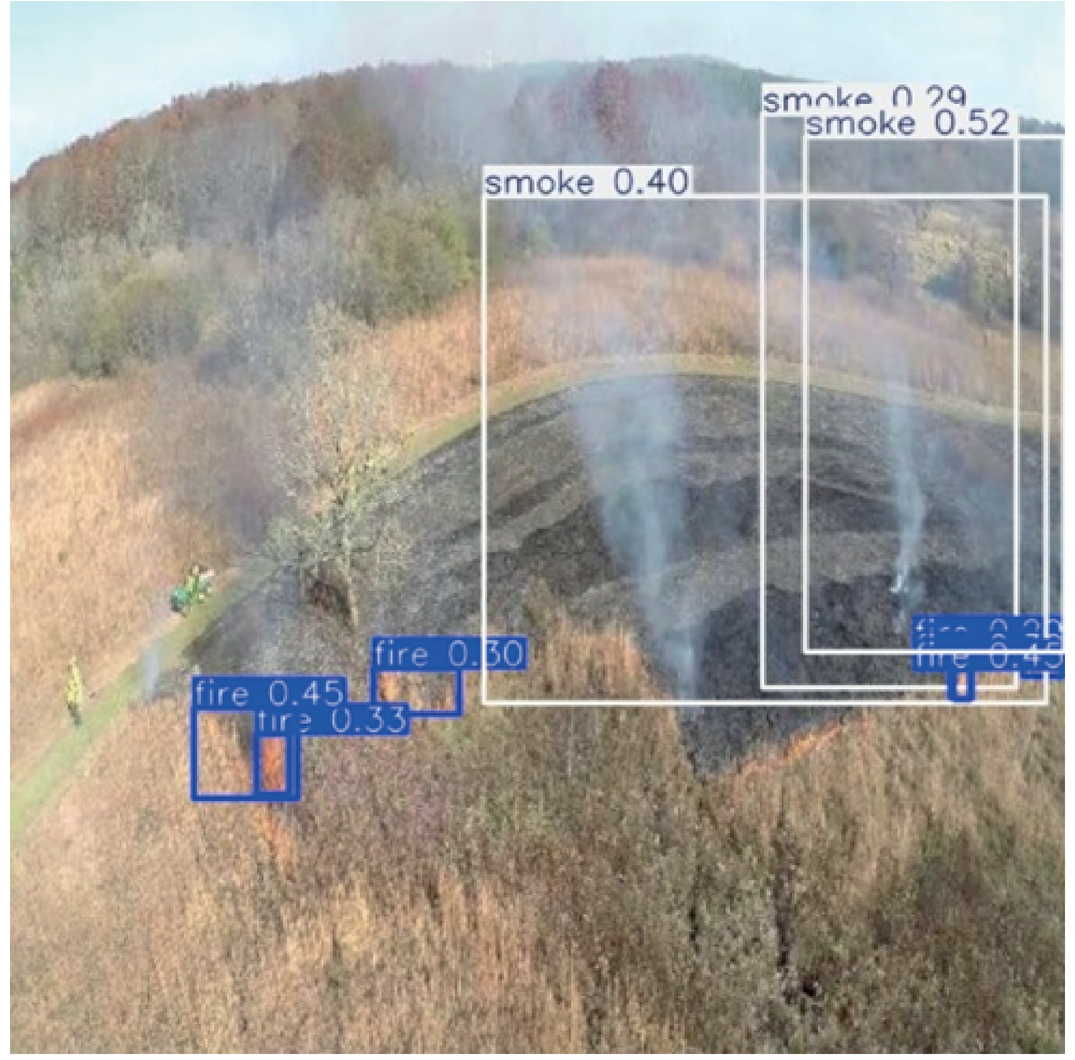
Performance Metrics



performance on the test dataset

Visual examples

Sample detection input
(fire and smoke bounding boxes)



Sample detection output
(fire and smoke bounding boxes)

AEC
Algerian Engineering Competition

Navigating Without GPS with SLAM

Our Solution

We integrate Visual SLAM (vSLAM), a foundational technology for true autonomous navigation.

System Architecture:

- The drone's Camera and IMU (Inertial Measurement Unit) data are fused and processed in real-time.
- The SLAM algorithm (like ORB-SLAM3) running on the onboard computer generates a 3D map of the environment.
- It simultaneously calculates the drone's precise position and orientation within that map, enabling fully autonomous navigation, obstacle avoidance, and mission execution without relying on GPS.

Project Feasibility: Architecture and Budget

Key Components:

Frame: Carbon Fiber Quadcopter (550–450 mm)

Flight Controller: Pixhawk 6C or Matek F765

Onboard Computer: Jetson Nano / Raspberry Pi 4

Cameras: FLIR Lepton & Raspberry Pi HQ

Compliance: The system architecture is designed for full compliance with Algerian regulations, specifically Decree 285-21, including registration, geovigilance, and electronic ID requirements.

Budget Estimate:

The total hardware cost for a single, fully-equipped drone prototype is estimated between 1,110$ and 1,250$ USD.

This demonstrates that our advanced, AI-powered solution is not only technically superior but also highly cost-effective and feasible for large-scale deployment.

The Aegis Project: The Future of Firefighting is Autonomous

Proven Solution:

Project Aegis presents a viable, transformative proof-of-concept. Our hybrid architecture directly solves the critical issue of response time and is built on a feasible, low-cost, and technically sound foundation.

Future Work & Vision:

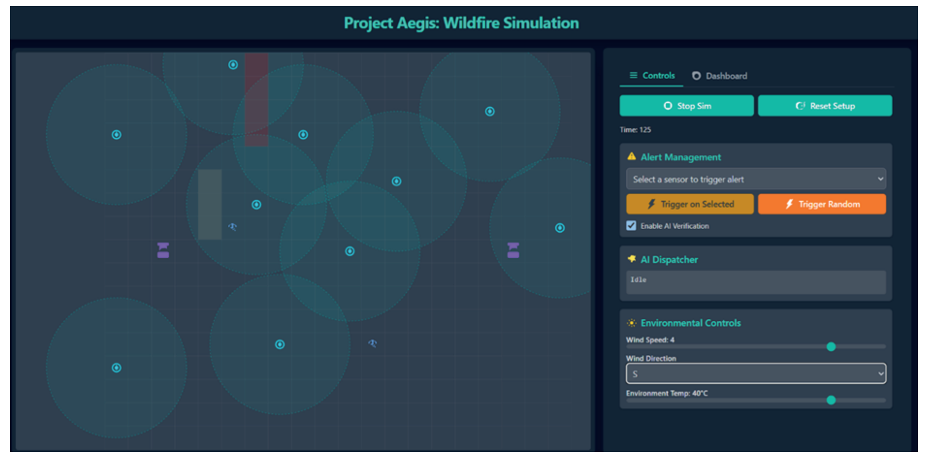
Hardware Integration & Field Testing : Moving from simulation to a physical prototype integrated with real LoRaWAN sensors

Predictive AI Models: Enhancing our risk mapping with predictive AI to anticipate fire behavior based on weather and environmental data.

Drone Swarm Coordination: Developing multi-drone collaboration algorithms for monitoring vast areas and coordinating complex verification and tracking missions

AEC
Algerian Engineering Competition

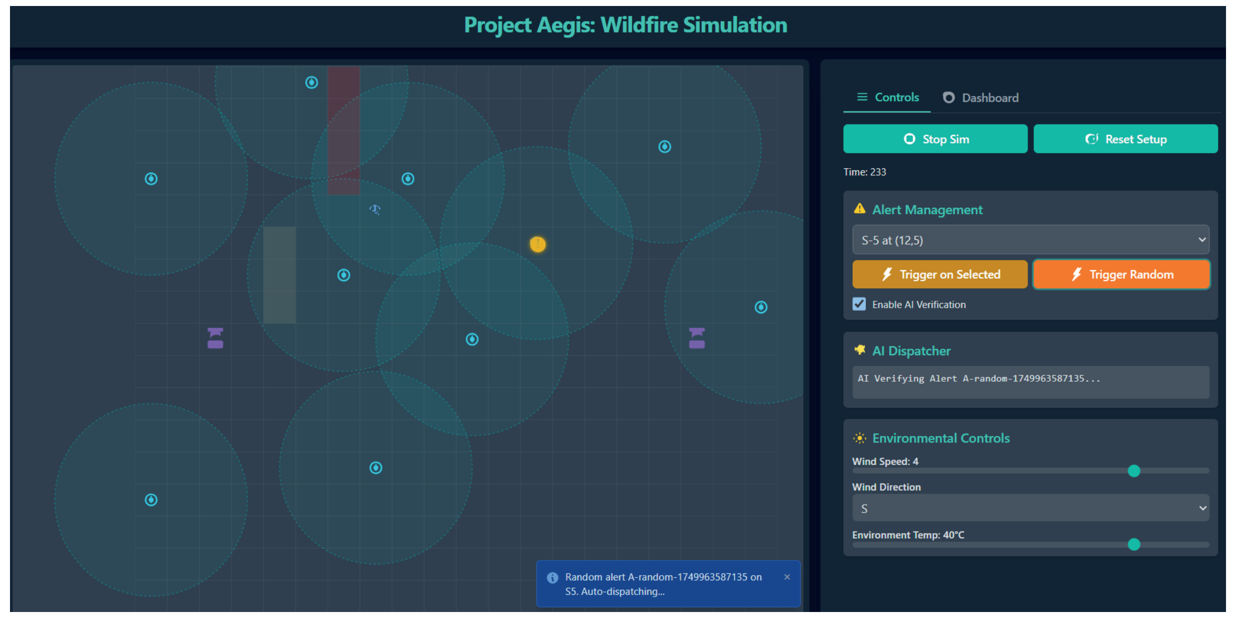
1 Forest Optimization Setup



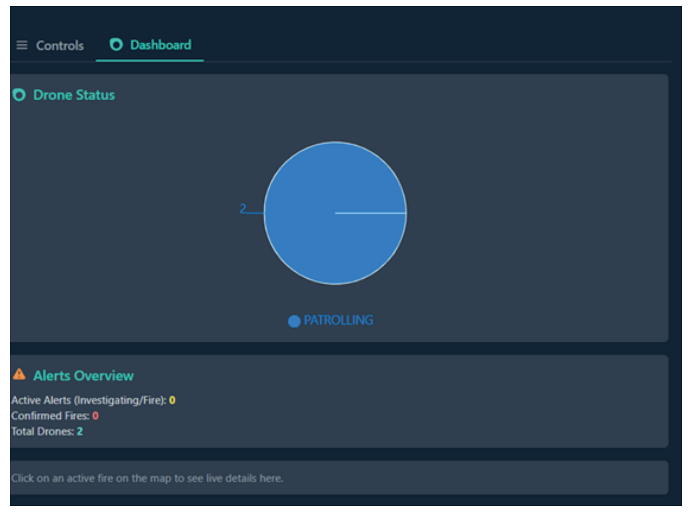
2 Simulation Engine

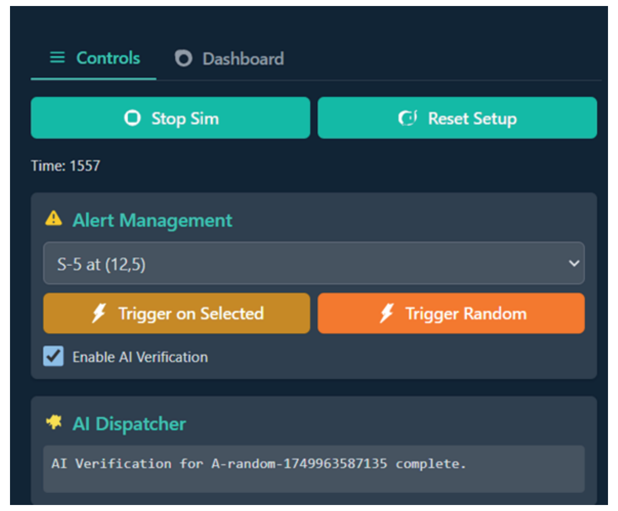
3 AI Integration

4 Control Panel



5 Dashboards & Modals





6 Real-Time Notifications

AEC
Algerian Engineering Competition

طاسيلي للعمل الجوي
Tassili Travail Aérien

Thank you

team robobo